IB MATH AA SL

Lesson 13

# Transformations of Graphs

Functions · 60-minute lesson

Syllabus SL 2.11 · 

# Do Now — recall

**ON YOUR WHITEBOARD**

1. Sketch roughly $y = x^2$ . Where is its lowest point?
2. What does adding 3 to a function do to its graph?
3. What does a minus sign in front, $-f(x)$, do?

**Answers (click to reveal)**

1. (0, 0)
2. shifts it UP 3
3. reflects it in the x-axis

# What we're learning today

## LEARNING OBJECTIVES

1. **Apply**
   vertical and horizontal translations.
2. **Apply**
   stretches (vertical and horizontal).
3. **Apply**
   reflections in the axes.
4. **Describe**
   a combined transformation precisely.

## SUCCESS CRITERIA — I CAN...

✓ translate a graph with $f(x)+k$ and $f(x-h)$

✓ stretch a graph with $a \cdot f(x)$ and $f(ax)$

✓ reflect a graph with $-f(x)$ and $f(-x)$

✓ describe transformations in the right order

# Translations

## SHIFTS

$$f(x)+k\ (up\ k)\ f(x-h)\ (¿\ h)$$

**Outside**

f(x) + k moves the graph UP by k

**Inside**

f(x − h) moves the graph RIGHT by h (note the sign!)

## SIGN TRAP

**f(x−2)**

moves RIGHT 2 (not left)

**f(x+2)**

moves LEFT 2

**Inside = opposite**

horizontal shifts feel 'backwards'.

# Describe a translation

**QUESTION**

Describe how the graph of $y = f(x)$ maps to

$y = f(x - 3) + 2$.

**SOLUTION**

Inside $(x - 3)$: horizontal shift RIGHT 3

Outside $(+ 2)$: vertical shift UP 2

So: translate 3 right and 2 up.

**Tip:**

*Inside the bracket = horizontal (and opposite sign); outside = vertical.*

# Your turn — transform

ANSWER IN YOUR BOOK

1. Describe $y = f(x) + 5$.
2. Describe $y = f(x + 4)$.
3. Describe $y = -f(x)$.

Answers (click to reveal)

1. up 5
2. left 4
3. reflection in the x-axis

# Stretches & reflections

## SCALING

$$a\ f(x)\ (\times a\ \ vertical)\ f(ax)\ (\times \frac{1}{a}\ \ horizontal)$$

**a·f(x)**

vertical stretch, scale factor a

**f(ax)**

horizontal stretch, scale factor 1/a

**Minus**

−f(x): reflect in x-axis;  f(−x): reflect in y-axis

## ORDER

Do stretches/reflections, then translations.

**Check**

a point: see where (1, f(1)) lands.

# Combined transformation

QUESTION

The graph of $y = f(x)$ is transformed to

$y = 2f(x) - 1$.

Describe the transformations.

SOLUTION

$2f(x)$: vertical stretch, scale factor 2

$-1$: translate DOWN 1

So: stretch ×2 vertically, then move down 1.

EXAM-STYLE QUESTION

# Exam-style question

The graph of $y = f(x)$ is transformed to $y = f(x + 2) - 3$.

(a) Describe fully the two transformations. [3]

(b) The point $(1, 4)$ is on $y = f(x)$. Find its image. [2]

MARK SCHEME — reveal after students attempt (tutor only)

(a) Translation 2 left (A1); translation 3 down (A1)(A1)

(b) $x: 1 \to -1$, $y: 4 \to 1$ (M1) $\to (-1, 1)$ (A1)

# Common mistakes

- **Inside sign**
  $f(x - 3)$ is RIGHT 3, not left 3.
- **Stretch vs translate**
  $a \cdot f(x)$ stretches; $f(x)+a$ shifts.
- **Horizontal scale**
  $f(ax)$ stretches by $1/a$, not $a$.
- **Reflection axis**
  $-f(x)$ reflects in x-axis; $f(-x)$ in y-axis.

# Mixed practice

**QUESTIONS**

1. Describe $y = f(x) - 4$.
2. Describe $y = 3\,f(x)$.
3. Describe $y = f(2x)$.
4. Describe $y = f(-x)$.

**Answers (click to reveal)**

1. down 4
2. vertical stretch ×3
3. horizontal stretch ×½
4. reflection in the y-axis

# Plenary & exit ticket

## KEY FORMULAS

$$f(x-h)+k:\ ¿\ h,\ up\ k$$

$$a\ f(x):\ stretch\ \times\ a$$

## EXIT TICKET — before you leave

1. Describe y = f(x) + 2.

2. Describe y = f(x − 5).

3. Describe y = −f(x).

**Answers (click to reveal)**

1. up 2
2. right 5
3. reflect in x-axis

Self-assess:

- need more help
- nearly there
- confident

# Homework

**SET ON DR FROST MATHS (auto-marked)**

**Task:**
Transformations of graphs.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

**NEXT LESSON**

**Lesson 14**

Quadratic functions & the discriminant — shapes, roots and the vertex.

$$= b^2 - 4ac$$